Chapter–3
Results

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Results

This chapter presents the data analysis findings pertaining to each of the four research aims.: (1) to identify the dominant typologies of everyday discrimination and the role of social categorization, (2) to examine how in-group and out-group identification predicting social distance towards members of out-group, (3) to assess intergenerational effects on social categorization and intergroup relationship and, (4) to understand the category salience and intergroup relationship in cross-categorization situation. The process of initial data processing for both qualitative and quantitative data types were examined, and the corresponding findings were presented. This practice enables readers to critically analyze the cognitive process underlying research findings and reduces the likelihood of introducing bias into the research methodology.

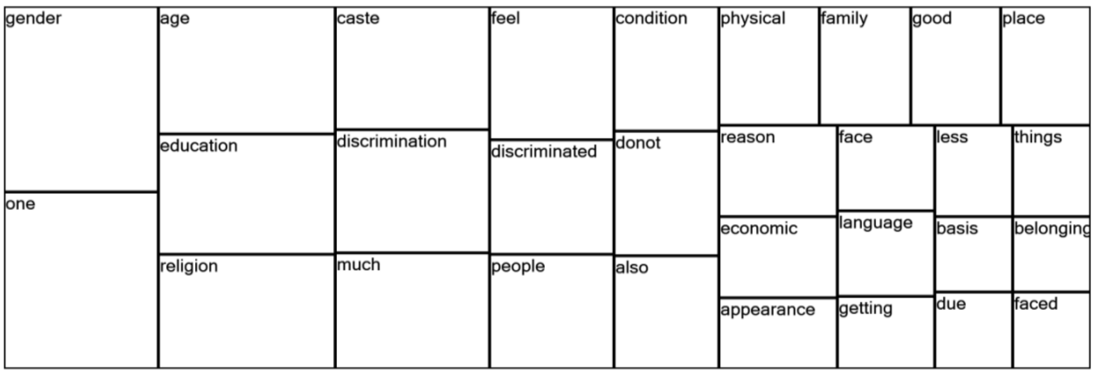
3.1 Phase – I

3.1.1 Aim 1: Dominant Typologies of Everyday Discrimination and the Role of Social Categorization

After organizing data into Excel spread sheet data was transferred in qualitative data analysis software program NVivo version 11+ developed by QSR International for the purpose of content analysis. Initially we conducted a word frequency query to get the most frequently occurring words or ideas in our dataset. Thirty most frequently occurring words were extracted from the source data as shown by hierarchical chart in figure 3.1.

Figure 3.1

Hierarchical Chart Obtained by Word Frequency Query Representing Thirty Frequently Occurring Words



This hierarchical chart is in the form of *tree map* which is used to analyse our hierarchies in one view and compare relative size of different aspects of data. Hierarchy charts use size of rectangles to convey meaning as we can see in figure 3.1, the size of rectangles are bigger for words which have higher occurrences than words having least occurrences. Based on result of word frequency query, Parent Nodes (Codes) were generated by the researcher with the help of eight steps provided by the Tesch (1990). A node can be defined as a compilation of references pertaining to a particular theme, case, or relationship. The collection of references can be accomplished by assigning sources to a designated node through the process of coding. Upon accessing the node, all the references are conveniently displayed in a centralized location. An initial categorization of source data into nodes was performed that yield 14 parent nodes that included 240 references. These fourteen parent node categories were age, gender, language, place, cast, socioeconomic status, race and ethnicity, achievement, religion, education, physical appearance, absence of discrimination, disability, and lifestyle as shown by hierarchical chart tree map in figure 3.2 and relative reference of each category in table 3.1. In this study these parent node categories were considered as typologies of daily discrimination faced by individuals. Extraction of these typologies addressed the first sub-question.

Figure 3.2

Hierarchical Chart Obtained by Referencing Cases to Parent Nodes

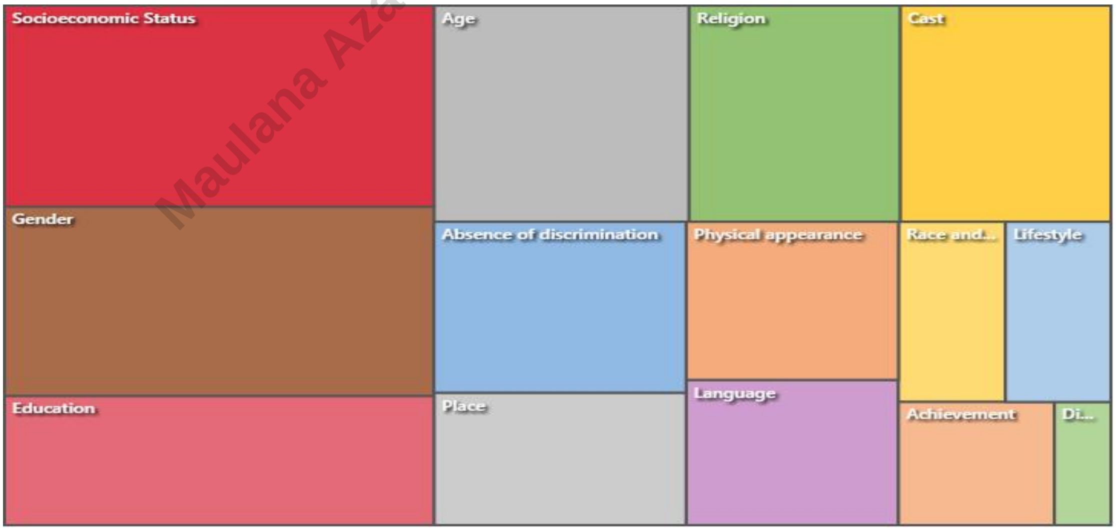


Table 3.1

Relative Reference Frequency of Nodes and Their Percentages

Node	Example quote	Relative referen f (%)
Age	No one helps me in my family, no one gives me money and food I am totally dependent upon pension so getting aged could be a factor due to which people don't want to be with me.	23(9.58)
Gender	Discriminated a lot based on gender at home and outside.	34(14.17)
Language	People make fun of me when I am unable to speak English and there is lot of discrimination on the basis of gender.	13(5.42)
Place	Have been felt discriminated on being Bihari (place) and on the basis of age.	14(5.83)
Cast	I have been discriminated against my cast and language many times at my school and workplace.	19(7.92)
Socioeconomic status	There is nothing too much but economic condition could be the reason for these experiences.	36(15.00)
Race and ethnicity	Family and ethnicity.	8(3.33)
Achievement	Discriminated against thinking and presumptions of people and also being highly professional.	8(3.33)
Religion	You can see Hindu-Muslim these days so religion could be one of the factor and children face issues in studies.	19(7.92)
Education	Age and education could be the reason of discrimination against me otherwise there is nothing like this.	23(9.58)
Physical appearance	I have Vitiligo (physical appearance) and also age could be the possible reason of discrimination against me.	14(5.83)
Absence of discrimination	I have less social contact so I do not face such issues.	18(7.50)
Disability	On my physical disability.	3(1.25)
Lifestyle	I have experienced discrimination on my type of work, the way we live and on my place of belonging.	8(3.33)

Note. Total references were 240.

To address the second sub-question of the study participants' demographic information was considered to see the relative distribution of generated references of typologies of everyday discrimination on their age, gender and area of living (urban/rural). Data was extracted from NVivo qualitative analysis software and bar graphs were constructed with the help of IBM SPSS version 28 as shown in figure 3.3, 3.4 and 3.5

Figure 3.3

Representing Relative Distribution of Generated References of Typologies of Discrimination Based on Participants' Gender

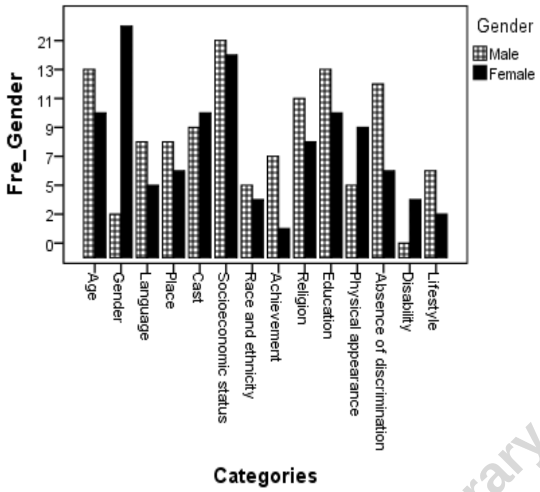


Figure 3.4

Representing Relative Distribution of Generated References of Typologies of Discrimination Based on Participants' Age (Young was 44 Years Below and Old was Above 44 Years (WHO,2015)

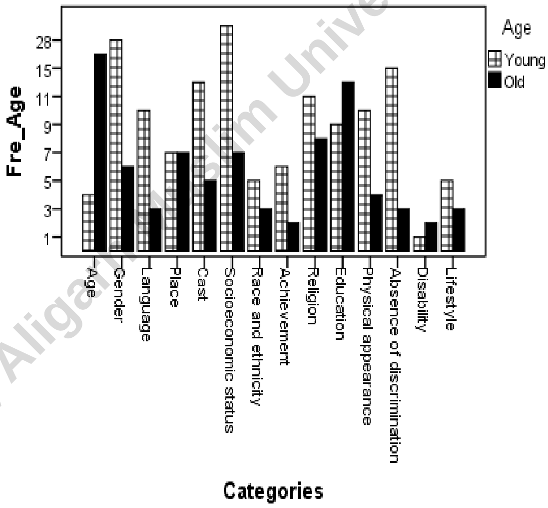
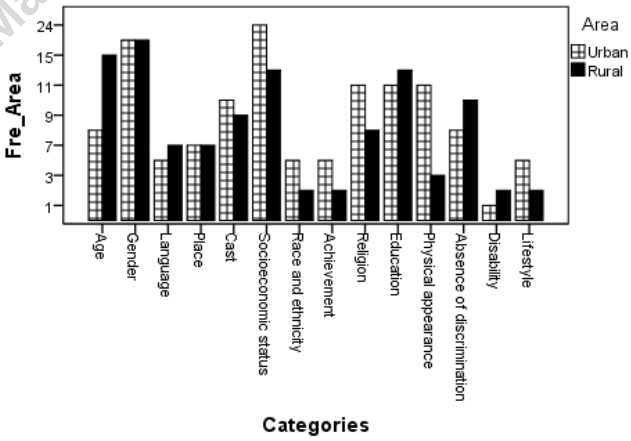


Figure 3.5

Representing Relative Distribution of Generated References of Typologies of Discrimination Based on Participants' Area of Living (Urban/Rural)



Referring to figure 3.3 greater numbers of male participants reported incidents of discrimination against them on their age, language, and place of belongingness, achievement, religion, education and lifestyle than female participants. Greater numbers of male participants also reported incidence of no discrimination than female participants. Additionally, greater number of female participants reported incidents of discrimination against them on their gender, physical appearance and disability than male participants whereas almost equal number of male and female participants reported incidents of discrimination against them on their cast, socioeconomic status, race and ethnicity.

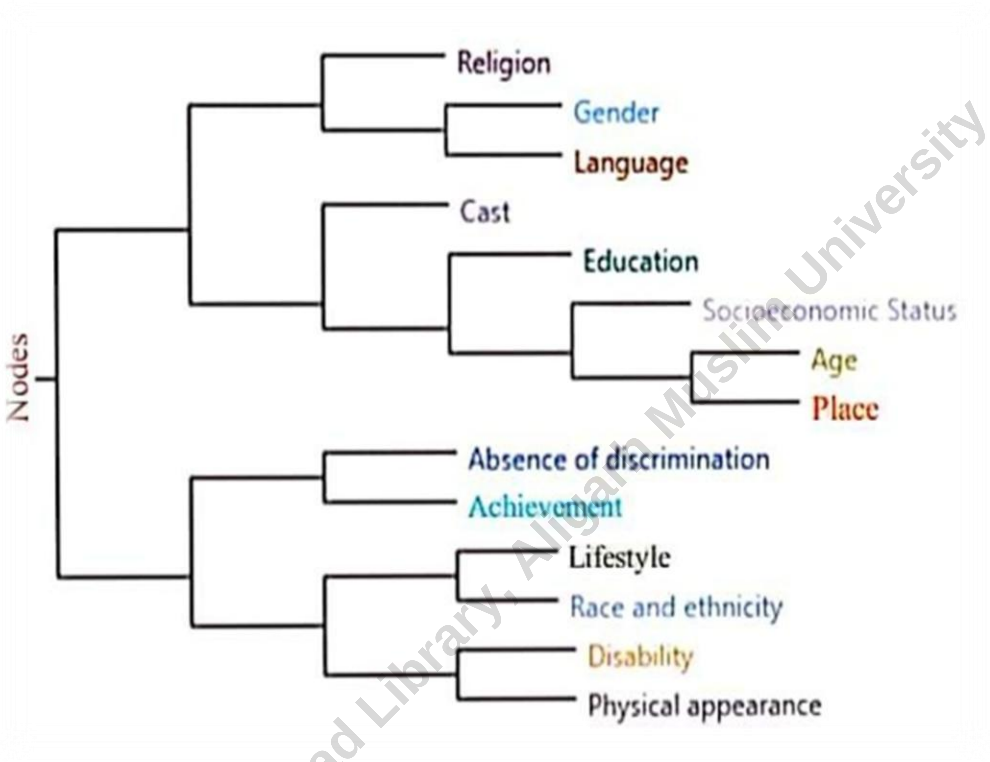
Talking about the age, figure 3.4 suggested that greater number of young people (below or equal to 44 years) were prone to face discrimination on their gender, language, cast, socioeconomic status, race and ethnicity, achievement, religion, physical appearance and lifestyle than older participants (above 44 years). Greater numbers of young participants also reported incidence of no discrimination than older participants. Older people reported more incidents of discrimination against them on their age, education and disability, whereas equal number of young and old participants reported incidents of discrimination against them on their place of living.

Figure 3.5, showing that participants who were living in urban area reported more incidents of discrimination against them on their socioeconomic status, race and ethnicity, achievement, religion, physical appearance and lifestyle than participants who were living in rural area. Individuals from rural area faced more discrimination then individuals from urban area on their age, education, disability, and they also reported more incidence of no discrimination. Both urban and rural participants reported almost equal incidents of discrimination against them on their gender, language, place of belonging, and cast.

Cluster analysis was carried out to address the third sub-question of this study. Cluster analysis is a method employed for exploratory purposes, enabling the visualization of patterns within data by grouping sources or nodes that exhibit similarities in terms of shared words, attribute values, or coding by nodes. Cluster analysis diagrams offer a visual depiction of sources or nodes, facilitating the identification of shared characteristics and distinctions. In the cluster analysis diagram, sources or nodes that exhibit proximity are indicative of a higher degree of similarity compared to those that are situated at a greater distance from each other. In this study, a cluster analysis diagram was constructed in the form of a horizontal dendrogram, as depicted in figure 3.6. The horizontal dendrogram is a branching diagram that arranges similar items on the same branch while placing dissimilar items further apart.

Figure 3.6

Nodes Clustered by Word Similarity



To cluster the nodes NVivo calculated the Jaccard similarity coefficient, it is a statistics used for gauging the similarity and diversity of nodes, and ranges from 0 to 1 (0 = least similar, 1 = most similar). Based on Jaccard's coefficient lifestyle and race and ethnicity node pair shared .28 ratio of intersection over union suggested 28% similarity, place and age node pair shared .21 ratio of intersection over union suggested 21% similarity, lifestyle and age education node pair shared .17 ratio of intersection over union suggested 17% similarity, place and language node pair shared .15 ratio of intersection over union suggested 15% similarity, and other node pairs fall below .15 ratio of intersection over union which means they shared least percentage of similarity and remained independent of each other. These findings manifested that different typologies of discrimination remained mutually exclusive.

3.2 Phase – II

Before the statistical analysis, data screening procedure was carried out to ensure that the data must be clean and satisfying the mandatory assumptions (normality, homoscedasticity, independence etc.) of parametric tests for aim 2, 3 and 4. A normality test was conducted for the data distribution in order to verify that there were no violations of the assumption of normality, which is a fundamental requirement for the application of inferential statistics (Chua, 2013). The normality of the data was assessed using the skewness and kurtosis indices in SPSS, as outlined by Pallant (2005, 2013). According to the literature (Byrne, 2010; Garson, 2012; Hair et al., 2010), it is generally accepted that data is considered to be normally distributed within the range of skewness values between -2 and +2, as well as kurtosis values between -7 and +7. Hoyle (1995) posits that when dealing with sample sizes exceeding 300, it is appropriate to use an absolute skewness value greater than two or an absolute kurtosis value greater than seven as benchmarks for identifying significant departures from normality. No variables within the study were found to exhibit abnormal distribution or were excluded from the initial procedure based on different preferred social categories. The range of skewness indices for the variables under investigation across various preferred social categories ranged from -1.73 to 1.51, while the range of kurtosis indices spanned from -4.140 to 3.602. Therefore, it can be inferred that the collected data exhibited a normal distribution, as indicated by the skewness and kurtosis indices of the variables in various preferred social categories falling within the range expected for a normal distribution. Additional assumptions were assessed through the utilization of appropriate statistical methods, specifically for parametric statistical tests. No assumptions were examined for the non-parametric statistical tests used in the study. Following data screening, hypotheses were examined using relevant statistical tests based on the nature of the data. Results were reported using the APA 7th edition Publication Manual (APA, 2020). Effect size for each statistics were calculated and reported as per *Journal Article Reporting Standards* (JARS) provide by *American Psychological Association* (APA, 2020). The descriptions of different effect size used in this study along with their range has been provided in Appendix–*I*.

3.2.1 Aim 2: In-Group and Out-Group Identification Predicting Social Distance

Table 3.2

Frequencies and Goodness of Fit Chi-Square Test for Social Categories in Terms of Their Preferences

Social category	f_o	%	χ^2 (7)	Cohen's ω
Age	42	5.83		
Gender	74	10.28		
Lingual	36	5.00		
Religion	172	23.89	251***	.59
Place	54	7.50		
Educational group	180	25.00		
Caste	61	8.47		
Economic group	101	14.03		

Note. N = 720. f_o was observed frequency. Expected frequency of each social category f_e was 90 (12.5%).
***p < .001.

In table 3.2, Goodness-of-Fit Chi-square test was performed to determine distribution of social categories across sample in terms of their preferences. Results revealed that preferences for the social categories were unequally distributed across sample, χ^2 (7, N = 720) = 251, p < .001, ω = .59. The value of Cohen's ω was .59 (> .50) which indicated large effect size. Hence, social categories followed hypothesized distribution as stated in hypothesis H_1 "*the distribution of social categories across the sample will be unequal in terms of their preferences*". Results supported hypothesis H_1 in the present study. Additionally, across sample, 25% participant's preferred educational group as their preferred social category which was twice than expected. 23.89%, nearly twice than expected, participants preferred religion as their preferred social category. 14.03% individuals reported economic group as their preferred social category whereas 10.28% people preferred gender, 8.47% preferred caste, 7.50% preferred place, 5.83% preferred age, and only 5% individuals identified themselves with their lingual category.

Table 3.3

Mean, Standard Deviation and Correlation Matrix for In-Group Identification, Out-Group Identification and Segments of Social Distance for Different Preferred Social Category

Variable	*n*	*M*	*SD*	Correlations 1	2	3	4
Age							
1. In-group identification	42	24.90	2.83	—			
2. Out-group identification	42	16.98	4.73	-.04	—		
3. Social distance intensity	42	38.00	11.76	.29	.06	—	
4. Perceived social distance	42	1.71	0.97	.26	-.05	.78***	—
Gender							
1. In-group identification	74	24.51	2.70	—			
2. Out-group identification	74	14.96	4.16	.20	—		
3. Social distance intensity	74	35.54	6.97	-.22	-.22	—	
4. Perceived social distance	74	1.20	0.62	-.01	-.18	.63***	—
Lingual							
1. In-group identification	36	26.25	1.73	—			
2. Out-group identification	36	12.42	4.54	-.10	—		
3. Social distance intensity	36	48.06	21.08	-.10	-.66***	—	
4. Perceived social distance	36	2.58	1.53	-.11	-.53***	.72***	—
Religion							
1. In-group identification	172	25.72	2.75	—			
2. Out-group identification	172	7.48	4.22	-.14	—		
3. Social distance intensity	172	70.16	28.40	.25***	-.30***	—	
4. Perceived social distance	172	3.96	1.74	.32***	-.26***	.82***	—
Place							
1. In-group identification	54	26.44	2.48	—			
2. Out-group identification	54	14.48	4.40	-.30*	—		
3. Social distance intensity	54	44.31	14.07	.17	-.38**	—	
4. Perceived social distance	54	2.11	1.00	.23	-.38**	.72***	—
Educational group							
1. In-group identification	180	25.39	2.82	—			
2. Out-group identification	180	13.54	5.86	-.003	—		
3. Social distance intensity	180	54.68	22.60	.16*	-.32***	—	
4. Perceived social distance	180	2.89	1.57	.18*	-.32***	.70***	—
Caste							
1. In-group identification	61	26.18	2.35	—			
2. Out-group identification	61	10.54	5.19	-.20	—		
3. Social distance intensity	61	58.84	22.68	.19	-.51***	—	
4. Perceived social distance	61	3.41	1.65	.18	-.58***	.64***	—
Economic group							
1. In-group identification	101	22.79	3.82	—			
2. Out-group identification	101	11.78	5.28	.21*	—		
3. Social distance intensity	101	45.77	16.30	.02	-.44***	—	
4. Perceived social distance	101	2.26	1.25	.001	-.50***	.75***	—

Note. N = 720.
*$p < .05$. **$p < .01$. ***$p < .001$.

The Pearson Product-Moment correlation coefficient was employed to measure whether there was a correlation between intergroup identification (in-group and out-group identification) and segments of social distance (perceived social distance and social distance intensity) on individuals' preferred social categories.

In-group identification exhibited a significant weak positive correlation with social distance intensity on religion (r = .25, p < .001) and education (r = .16, p = .030) as participants' preferred social category. On the other hand, in-group identification had an insignificant weak positive correlation with social distance intensity on age (r = .29, p = .061), place (r = .17, p = .228), and caste (r = .19, p = .152) as participants' preferred social category. Also, in-group identification had an insignificant weak negative correlation with social distance intensity on gender (r = -.22, p = .054) and language (r = -.10, p = .587) as participants' preferred social category. Additionally, there was a negligible correlation between in-group identification and the social distance intensity on the economic group (r = .02, p = .803) as participants' preferred social category. In-group identification and social distance intensity shared 6.25% (r^2 = .0625) of the variance on religion and 2.56% (r^2 = .0256) of the variance on education as participants' preferred social categories.

In-group identification had a significant weak positive correlation with perceived social distance on religion (r = .32, p < .001) and education (r = .18, p = .017) as participants' preferred social category, whereas in-group identification had an insignificant weak positive correlation with perceived social distance on age (r = .26, p = .102), place (r = .23, p = .095), and caste (r = .18, p = .168) as participants' preferred social category; in-group identification had an insignificant weak negative correlation with perceived social distance on language (r = -.10, p = .523) as participants' preferred social category. Nevertheless, in-group identification was negligibly associated with perceived social distance on gender (r = -.01, p = .961) and the economic group (r = .001, p = .993) as participants' preferred social category. In-group identification and perceived social distance shared 10.24% (r^2 = .1024) of the variance on religion and 3.24% (r^2 = .0324) of the variance on education as participants' preferred social categories.

Out-group identification had a significant strong negative correlation with social distance intensity on the lingual (r = -.66, p < .001) social category; and a significant moderate negative correlation on place (r = -.38, p = .005), education (r = -

.32, $p < .001$), caste ($r = -.51$, $p < .001$), and the economic group ($r = -.44$, $p < .001$) as participants' preferred social category; conversely, out-group identification had a significant weak negative relationship with social distance intensity on religion ($r = -.30$, $p < .001$) as participants' preferred social category; out-group identification had an insignificant weak negative correlation with social distance intensity on gender ($r = -.22$, $p = .054$) as participants' preferred social category and there was a negligible correlation between out-group identification and the social distance intensity on age ($r = .06$, $p = .681$) . Out-group identification and social distance intensity shared 43.56% ($r^2 = .4356$) of the variance in language, 14.44% ($r^2 = .1444$) of the variance in place, 10.24% ($r^2 = .1024$) of the variance in education, 26.01% ($r^2 = .2601$) of the variance in caste, 19.36% ($r^2 = .1936$) of the variance in the economic group, and 9.0% ($r^2 = .090$) of the variance in religion as participants' preferred social category.

Out-group identification had a significant moderate negative correlation with perceived social distance on lingual ($r = -.53$, $p < .001$), place ($r = -.38$, $p = .004$), education ($r = -.32$, $p < .001$), caste ($r = -.58$, $p < .001$), and economic group ($r = -.50$, $p < .001$) as participants' preferred social category; out-group identification had a significant weak negative correlation with perceived social distance on religion ($r = -.26$, $p < .001$) as participants preferred social category; out-group identification had an insignificant weak negative correlation with perceived social distance on gender ($r = -.18$, $p = .130$) as participants' preferred social category; out-group identification was negligibly related to perceived social distance on age ($r = -.05$, $p = .731$) as participants' preferred social category. Out-group identification and perceived social distance shared 28.09% ($r^2 = .2809$) of the variance on lingual, 14.44% ($r^2 = .1444$) of the variance on place, 10.24% ($r^2 = .1024$) of the variance on education, 33.64% ($r^2 = .3364$) of the variance on caste, 25.0% ($r^2 = .250$) of the variance on economic group, and 6.76% ($r^2 = .0676$) of the variance on religion as participants' preferred social category.

In-group and out-group identification exhibited significant weak negative correlation with the place ($r = -.30$, $p = .026$) and a significant weak positive correlation on economic group ($r = .21$, $p = .038$) as the preferred social category of the participants. In-group and out-group identification had an insignificant weak negative correlation with lingual ($r = -.10$, $p = .587$), religion ($r = -.14$, $p = .057$), and caste (r = -.20, p = .139); and an insignificant weak positive correlation on gender (r = .20, p = .091) as the preferred social category of the participants. In-group and out-group identification were negligibly correlated on age (r = -.04, p = .801) and education (r = -.003, p = .964) as participants' preferred social category. In-group and out-group identification shared 9.0% (r^2 = .090) of the variance on place and 4.41% (r^2 = .0441) of the variance on economic group.

Social distance intensity and perceived social distance had a strong significant positive correlation on each social category, age (r = .78, p < .001), gender (r = .63, p < .001), lingual (r = .72, p < .001), religion (r = .82, p < .001), place (r = .72, p < .001), education (r = .70, p < .001), caste (r = .64, p < .001), and economic group (r = .75, p < .001) as preferred by individuals, and shared 60.84% (r^2 = .6084) of the variance on age, 39.69% (r^2 = .3969) of the variance on gender, 51.84% (r^2 = .5184) of the variance on lingual, 67.24% (r^2 = .6724) of the variance on religion, 51.84% (r^2 = .5184) of the variance on place, 49.0% (r^2 = .490) of the variance on education, 40.96% (r^2 = .4096) of the variance on caste, and 56.25% (r^2 = .5625) of the variance on economic group. Based on findings hypotheses H_{2a} "*there will be relationship between in-group identification, out-group identification, and social distance intensity in participant' preferred social category*" and H_{2b} "*there will be relationship between in-group identification, out-group identification, and perceived social distance in participants' preferred social category*" were partially supported.

Table 3.4

Standard Multiple Linear Regression Analysis for In-Group and Out-Group Identification Predicting Social Distance Intensity

Predictor	*B*	*SEB*	β	*t*	R^2	*adj.* R^2	Cohen's f^2	95% *CI* of *B* [*LL*,*UL*]
Age ($F(2, 39) = 1.94$, $p = .157$)								
Constant	4.35	17.38		.250				[-30.80,39.50]
X_1	1.22	0.63	.29	1.92	.091	.044	0.10	[-0.06,2.50]
X_2	0.19	0.38	.08	0.50				[-0.58,0.96]
Gender ($F(2, 71) = 3.19$*, $p = .047$)								
Constant	51.85	7.41		7.00***				[37.07,66.62]
X_1	-0.47	0.30	-.18	1.57	.082	.057	0.09	[-1.07,0.13]
X_2	-0.32	0.19	-.19	1.63				[-0.70,0.07]
Lingual ($F(2, 33) = 14.04$***, $p < .001$)								
Constant	136.80	42.54		3.22**				[50.26,223.34]
X_1	-1.90	1.57	-.16	1.21	.460	.427	0.85	[-5.09,1.27]
X_2	-3.13	0.60	-.68	5.25***				[-4.34,-1.92]
Religion ($F(2, 169) = 13.60$***, $p < .001$)								
Constant	26.81	20.13		1.33				[-12.93,66.55]
X_1	2.22	0.75	.22	2.98**	.139	.128	0.16	[0.75,3.70]
X_2	-1.84	0.48	-.27	3.80***				[-2.80,-0.87]
Place ($F(2, 51) = 4.40$*, $p = .017$)								
Constant	52.60	23.10		2.28*				[6.25,98.96]
X_1	0.32	0.77	.06	0.42	.147	.114	0.17	[-1.22,1.86]
X_2	-1.16	0.43	-.36	2.67**				[-2.03,-0.29]
Educational group ($F(2, 177) = 13.01$, $p < .001$)								
Constant	38.65	14.82		2.61**				[9.40,67.90]
X_1	1.29	0.56	.16	2.29*	.128	.118	0.15	[0.18,2.40]
X_2	-1.23	0.27	-.32	4.55***				[-1.76,-0.70]
Caste ($F(2, 58) = 10.80$, $p < .001$)								
Constant	58.72	30.43		1.93				[-2.18,119.63]
X_1	0.88	1.10	.09	0.80	.271	.246	0.37	[-1.33,3.08]
X_2	-2.17	(0.50)	-.50	4.34***				[-3.17,-1.17]
Economic group ($F(2, 98) = 12.50$, $p < .001$)								
Constant	50.80	9.00		5.64***				[32.94,68.66]
X_1	0.51	0.40	.12	1.30	.203	.187	0.25	[-0.27,1.29]
X_2	-1.42	0.28	-.46	4.99***				[-1.98,-0.85]

Note. X_1 was In-group identification and X_2 was Out-group identification. X_1 and X_2 were predictors and social distance intensity was criterion.
*$p < .05$. **$p < .01$. ***$p < .001$.

To assess violations of linear assumptions, various techniques were employed. Residuals analysis was conducted to visually examine heteroscedasticity. Additionally, variance inflation factors (VIF) and tolerance statistics were utilized to identify Multicollinearity, specifically by considering VIF values exceeding 10 and tolerance values below 0.1 shows Multicollinearity. Furthermore, the Durbin-Watson (DW) test was employed to detect non-independence of errors, with values below 2 and above DW indicating such non-independence (Field, 2009). Q-Q plots suggested normal distribution of variables. The Durbin-Watson statistic, with an average value of 1.58, suggests the absence of serial correlations among errors. All variance inflation factors (VIF) exhibited values below 10, while the average VIF surpassed 1. Nevertheless, none of the tolerance statistics fell below 0.2, suggesting that the presence of Multicollinearity was improbable. Table 3.4 shows the extent` to which social distance intensity predicted by in-group identification (X_1) and out-group identification (X_2) in each social category preferred by participants' as their preferred social category. For *t* statistical value in table 3.4 we mentioned strength not the direction, direction of predictors could be interpreted by the direction of *B* or β.

For *Age* as preferred social category, the R^2 value of .091 revealed that the predictors (X_1 and X_2) explained 9.1% variance in the social distance intensity score with $F(2, 39) = 1.94$, $p = .157$, $f^2 = 0.10$. The calculated Cohen's f^2 was 0.10 which suggested small effect size. The findings pointed out that in-group identification ($\beta = .29$, $p = .061$) and out-group identification ($\beta = .08$, $p = .616$) emerged as non-significant predictors of social distance intensity.

For *Gender* as preferred social category, the R^2 value of .082 revealed that the predictors (X_1 and X_2) explained 8.2% variance in the social distance intensity score with $F(2, 71) = 3.19$, $p = .047$, $f^2 = 0.09$. The calculated value of Cohen's f^2 was 0.09 which implied small effect size. The findings demonstrated that in-group identification ($\beta = -.18$, $p = .121$) and out-group identification ($\beta = -.19$, $p = .108$) emerged as insignificant predictors of social distance intensity.

For *Lingual* as preferred social category, the R^2 value of .460 revealed that the predictors (X_1 and X_2) explained 46.0% variance in the social distance intensity score with $F(2, 33) = 14.04$, $p < .001$, $f^2 = 0.85$. The obtained value of Cohen's f^2 was 0.85 which pointed out being large effect size. The results shown that out-group identification negatively predicted social distance intensity ($\beta = -.68$, $p < .001$) whereas in-group identification emerged as a non-significant predictor of social distance intensity ($\beta = -.16$, $p = .234$).

For *Religion* as preferred social category, the R^2 value of .139 revealed that the predictors (X_1 and X_2) explained 13.9% variance in the social distance intensity score with $F(2, 169) = 13.60$, $p < .001$, $f^2 = 0.16$. The Cohen's f^2 was 0.16 which suggesting medium effect size. The results highlighted that in-group identification positively predicted social distance intensity (β = .22, p = .003) whereas out-group identification emerged as a negative predictor of social distance intensity (β = -.27, p < .001).

For *Place* as preferred social category, the R^2 value of .147 revealed that the predictors (X_1 and X_2) explained 14.7% variance in the social distance intensity score with $F(2, 51) = 4.40$, $p = .017$, $f^2 = 0.17$. The Cohen's f^2 was 0.17 which expressed medium effect size. The findings reported that out-group identification negatively predicted social distance intensity (β = -.36, p = .01) whereas in-group identification emerged as an insignificant predictor of social distance intensity (β = .06, p = .676).

For *Education* as preferred social category, the R^2 value of .128 revealed that the predictors (X_1 and X_2) explained 12.8% variance in the social distance intensity score with $F(2, 177) = 13.01$, $p < .001$, $f^2 = 0.15$. The calculated Cohen's f^2 was 0.15 which demonstrated medium effect size. The results pointed out that in-group identification positively predicted social distance intensity (β = .16, p = .023) whereas out-group identification emerged as a negative predictor of social distance intensity (β = -.32, p < .001).

For *Caste* as preferred social category, the R^2 value of .271 revealed that the predictors (X_1 and X_2) explained 27.1% variance in the social distance intensity score with $F(2, 58) = 10.80$, $p < .001$, $f^2 = 0.37$. The Cohen's f^2 was 0.37 which implied large effect size. The findings demonstrated that out-group identification negatively predicted social distance intensity (β = -.50, p < .001) whereas in-group identification emerged as an insignificant predictor of social distance intensity (β = .09, p = .430).

For *Economic group* as preferred social category, the R^2 value of .203 revealed that the predictors (X_1 and X_2) explained 20.3% variance in the social distance intensity score with $F(2, 98) = 12.50$, $p < .001$, $f^2 = 0.25$. The obtained Cohen's f^2 was 0.25 which suggested medium effect size. The findings clarified that out-group identification negatively predicted social distance intensity (β = -.46, p < .001) whereas in-group identification emerged as a non-significant predictor of social distance intensity (β = .12, p = .195). Results revealed that hypothesis H_3 "*regression coefficient to predict social distance intensity based on in-group and out-group identification will be other than zero in participants' preferred social category*" was supported for religion and educational group; partially supported for lingual, place, caste and economic group; did not supported for age and gender as participants' preferred social category.

Table 3.5

Standard Multiple Linear Regression Analysis for In-Group and Out-Group Identification Predicting Perceived Social Distance

Predictor	*B*	*SEB*	β	*t*	R^2	*adj.* R^2	Cohen's f^2	95% *CI of B* [*LL*,*UL*]
Age (*F*(2, 39) = 1.41, *p* = .256)								
Constant	-0.30	1.45		0.20				[-3.23,2.64]
X_1	0.09	0.05	.25	1.64	.067	.020	0.07	[-0.02,0.19]
X_2	-0.01	0.03	-.04	0.29				[-0.07,0.06]
Gender (*F*(2, 71) = 1.20, *p* = .310)								
Constant	1.44	0.68		2.13*				[0.09,2.79]
X_1	0.01	0.03	.03	0.26	.032	.005	0.03	[-0.05,0.06]
X_2	-0.03	0.02	-.18	1.54				[-0.06,0.01]
Lingual (*F*(2, 33) = 7.17**, *p* = .003**)**								
Constant	8.61	3.52		2.44*				[1.44,15.78]
X_1	-0.14	0.13	-.16	1.10	.303	.261	0.43	[-0.41,0.12]
X_2	-0.18	0.05	-.54	3.71***				[-0.28,-0.08]
Religion (*F*(2, 169) = 14.69***, *p* < .001)								
Constant	0.01	1.23		0.01				[-2.42,2.42]
X_1	0.18	0.04	.28	3.97***	.148	.138	0.17	[0.09,0.27]
X_2	-0.09	0.04	-.22	3.07**				[-0.15,-0.03]
Place (*F*(2, 51) = 4.93*, *p* = .011)								
Constant	1.93	1.63		1.18				[-1.34,5.21]
X_1	0.05	0.05	.12	0.92	.162	.129	0.19	[-0.06,0.16]
X_2	-0.08	0.03	-.35	2.58*				[-0.14,-0.02]
Educational group (*F*(2, 177) = 13.24***, *p* < .001)								
Constant	1.54	1.03		1.50				[-0.50,3.57]
X_1	0.10	0.04	.18	2.51*	.130	.120	0.15	[0.02,0.18]
X_2	-0.08	0.02	-.31	4.48***				[-0.12,-0.05]
Caste (*F*(2, 58) = 14.75***, *p* < .001)								
Constant	4.00	2.11		1.89				[-0.23,8.21]
X_1	0.05	0.08	.07	0.65	.337	.314	0.51	[-0.10,0.20]
X_2	-0.18	0.04	-.56	5.17***				[-0.25,-0.11]
Economic group (*F*(2, 98) = 17.06***, *p* < .001)								
Constant	2.90	0.66		4.36***				[1.58,4.21]
X_1	0.04	0.03	.11	1.22	.258	.243	0.35	[-0.02,0.09]
X_2	-0.12	0.02	-.52	5.84***				[-0.16,-0.08]

Note. X_1 was In-group identification and X_2 was Out-group identification. X_1 and X_2 were predictors and perceived social distance was criterion.
**p* < .05. ***p* < .01. ****p* < .001.

To assess violations of linear assumptions, various techniques were employed. Residuals analysis was conducted to visually examine heteroscedasticity. Additionally, variance inflation factors (VIF) and tolerance statistics were utilized to identify Multicollinearity, specifically by considering VIF values exceeding 10 and tolerance values below 0.1 shows Multicollinearity. Furthermore, the Durbin-Watson

(DW) test was employed to detect non-independence of errors, with values below 2 and above DW indicating such non-independence (Field, 2009). Q-Q plots suggested normal distribution of variables. The Durbin-Watson statistic, with an average value of 1.88, suggests the absence of serial correlations among errors. All variance inflation factors (VIF) exhibited values below 10, while the average VIF surpassed 1. Nevertheless, none of the tolerance statistics fell below 0.2, suggesting that the presence of Multicollinearity was improbable. For *t* statistical value in table 3.5 we mentioned strength not the direction, direction of predictors could be interpreted by the direction of *B* or β.

Table 3.5 shows the extent` to which perceived social distance predicted by in-group identification (X_1) and out-group identification (X_2) in each social category preferred by participants as their preferred social category.

For *Age* as preferred social category, the R^2 value of .067 revealed that the predictors (X_1 and X_2) explained 6.7% variance in the perceived social distance with $F(2, 39) = 1.41$, $p = .256$, $f^2 = 0.07$. The Cohen's f^2 was 0.07 which suggested small effect size. The results displayed that in-group identification (β = .25, $p = .109$) and out-group identification (β = -.04, $p = .775$) emerged as insignificant predictors of perceived social distance.

For *Gender* as preferred social category, the R^2 value of .032 revealed that the predictors (X_1 and X_2) explained 3.2% variance in the perceived social distance with $F(2, 71) = 1.20$, $p = .310$, $f^2 = 0.03$. The calculated Cohen's f^2 was 0.03 interpreted as small effect size. The results suggested that in-group identification (β = .03, $p = .798$) and out-group identification (β = -.18, $p = .127$) emerged as non-significant predictors of perceived social distance.

For *Lingual* as preferred social category, the R^2 value of .303 revealed that the predictors (X_1 and X_2) explained 30.3% variance in the perceived social distance with $F(2, 33) = 7.17$, $p = .003$, $f^2 = 0.43$. The obtained Cohen's f^2 was 0.43 which was considered having large effect size. The observations pointed out that out-group identification negatively predicted perceived social distance (β = -.54, $p < .001$) whereas in-group identification emerged as a non-significant predictor of perceived social distance (β = -.16, $p = .279$).

In *Religion* as preferred social category, the R^2 value of .148 revealed that the predictors (X_1 and X_2) explained 14.8% variance in the perceived social distance with $F(2, 169) = 14.69$, $p < .001$, $f^2 = 0.17$. The Cohen's f^2 was 0.17 which suggested medium effect size. The results suggested that in-group identification positively predicted perceived social distance (β = .28, $p < .001$) whereas out-group identification emerged as a negative predictor of perceived social distance (β = -.22, $p = .002$).

For *Place* as preferred social category, the R^2 value of .162 revealed that the predictors (X_1 and X_2) explained 16.2% variance in the perceived social distance with $F(2, 51) = 4.93$, $p = .011$, $f^2 = 0.19$. The calculated value of Cohen's f^2 was 0.19 which implied medium effect size. The outcomes suggested that out-group identification negatively predicted perceived social distance (β = -.35, $p = .013$) whereas in-group identification emerged as a non-significant predictor of perceived social distance (β = .12, $p = .360$).

For *Education* as preferred social category, the R^2 value of .130 revealed that the predictors (X_1 and X_2) explained 13.0% variance in the perceived social distance with $F(2, 177) = 13.24$, $p < .001$, $f^2 = 0.15$. The Cohen's f^2 was 0.15 which suggested medium effect size. The results demonstrated that in-group identification positively predicted perceived social distance (β = .18, $p = .013$) whereas out-group identification emerged as a negative predictor of perceived social distance (β = -.31, $p < .001$).

For *Caste* as preferred social category, the R^2 value of .337 revealed that the predictors (X_1 and X_2) explained 33.7% variance in the perceived social distance with $F(2, 58) = 14.75$, $p < .001$, $f^2 = 0.51$. The obtained Cohen's f^2 was 0.51 which referred as large effect size. The results pointed out that out-group identification negatively predicted perceived social distance (β = -.56, $p < .001$) whereas in-group identification emerged as a non-significant predictor of perceived social distance (β = .07, $p = .517$).

In *Economic group* as preferred social category, the R^2 value .258 revealed that the predictors (X_1 and X_2) explained 25.8% variance in the perceived social distance with $F(2, 98) = 17.06$, $p < .001$, $f^2 = 0.35$. The obtained Cohen's f^2 was 0.35 which suggested large effect size. The findings indicated that out-group identification negatively predicted perceived social distance (β = -.52, p < .001) whereas in-group identification emerged as a non-significant predictor of perceived social distance (β = .11, p = .226). Based on results hypothesis H_4 "*regression coefficient to predict perceived social distance based on in-group and out-group identification will be other than zero in participants' preferred social category*" was supported for religion and educational group; partially supported for lingual, place, caste and economic group; did not supported for age and gender as participants' preferred social category.

3.2.2 Aim 3: Intergenerational Effects on Social Categorization and Intergroup Relationship

Table 3.6

Mean, Standard Deviation and One-Way Analysis of Variance for In-Group Identification, Out-Group Identification and Social Distance Intensity Across Participants Preferred Social Category

Social category	In-group identification		Out-group identification		Social distance intensity	
	M	*SD*	*M*	*SD*	*M*	*SD*
Age	24.90_{ab}	2.83	16.98_{a}	4.73	38.00_{c}	11.76
Gender	24.51_{b}	2.69	14.96_{a}	4.16	35.54_{c}	6.97
Lingual	26.25_{a}	1.73	12.42_{bc}	4.54	48.06_{bc}	21.08
Religion	25.72_{a}	2.75	7.48_{d}	4.22	70.16_{a}	28.40
Place	26.44_{a}	2.48	14.48_{ab}	4.39	44.31_{c}	14.07
Educational group	25.39_{a}	2.82	13.54_{b}	5.87	54.68_{b}	22.60
Caste	26.18_{a}	2.35	10.54_{c}	5.19	58.84_{b}	22.68
Economic group	22.79_{c}	3.82	11.78_{c}	5.28	45.77_{c}	16.29
	F (7, 712) = 15.03*** η^2 = .13.		F (7, 712) = 35.37*** η^2 = .26.		F (7, 712) = 30.11*** η^2 = .23.	

Note. N = 720. Means in columns with different subscripts differ at the p = .05 level by Bonferroni (MCT).
***p < .001.

Table 3.6 shows mean, standard deviation, and F-values for in-group identification, out-group identification and social distance intensity across participants' highly preferred social category. Results indicated significant mean difference across participants' preferred social category on in-group identification F (7, 712) = 15.03, p < .001, η^2 = .13. The value of η^2 indicated medium effect size. Participants who preferred place (M = 26.44, SD = 2.48) as their preferred social category identified more with the members of their in-group (people belonging to same place) as compared to participants' who preferred other social category; and participants who preferred economic group (*M* = 22.79, *SD* = 2.48) as their preferred social category identified least with the members of their in-group (people belonging to similar economic condition as the participant).

Results shown significant mean difference across participants' preferred social category on out-group identification *F* (7, 712) = 35.37, *p* < .001, η^2 = .26. The value of η^2 indicated large effect size. Participants who preferred Age (*M* = 16.98, *SD* = 4.73) as their preferred social category identified more with the members of their out-group (people belonging to different age group as of participant) then compared to participants' who preferred other social category; and participants who preferred religion (*M* = 7.48, *SD* = 4.22) as their preferred social category identified least with the members of their out-group (people belonging to different religious orientation as of participant).

Results indicated significant mean difference across participants' preferred social category on their social distance intensity *F* (7, 712) = 30.11, *p* < .001, η^2 = .23. The value of η^2 indicated large effect size. Participants who preferred religion (*M* = 70.16, *SD* = 28.40) as their preferred social category exhibited higher intensity of social distance towards members of their out-group (people belonging to different religious orientation) as compared to participants' who preferred other social category; and participants who preferred gender (*M* = 35.54, *SD* = 6.97) as their preferred social category exhibited lower intensity of social distance towards members of their out-group (people belonging to different age group as of participant). Based on findings, our proposed hypothesis H_5 "*there will be mean differences on in-group identification, out-group identification, and social distance intensity across participants' preferred social category in whole sample*" was supported.

Table 3.7

Kruskal-Wallis Rank Sum Test for Perceived Social Distance Towards Members of Out-group Across Sample in Their Highly Preferred Social Category

Social category	N	M_{rank}	K-W $\chi^2(7)$	ε^2
Age	42	223.65		
Gender	74	135.78		
Lingual	36	342.36		
Religion	172	498.32		
Place	54	290.90	215.08***	.30
Educational group	180	381.28		
Caste	61	443.11		
Economic group	101	304.10		
Total	720			

*** $p < .001$.

The data were measured on ordinal scale for perceived social distance, therefore, in table 3.7 Kruskal-Wallis (K-W) rank sum test was carried out to assess if there was significant difference on perceived level of social distance towards members of out-group across participants' preferred social category. Higher mean ranks indicated higher perceived social distance towards embers of out-group. From table 3.7, outcome of the Kruskal-Wallis test was found to be significant, $\chi^2(7, N = 720) = 215.08$, $p < .001$, $\varepsilon^2 = .30$. The value of ε^2 was .30 (> .14) which indicated large effect size. Result of K-W test indicated that the M_{rank} of perceived social distance towards members of out-group was significantly varied across social categories preferred by participants as their preferred social category. Overall, individuals who preferred religion as their preferred social category expressed higher perceived social distance (M_{rank} = 498.32) towards members of out-group (if participant was religious then out group was non-religious and if participant was non-religious then out-group was religious), whereas individuals who preferred gender as their preferred social category expressed least perceived social distance (M_{rank} = 135.78) towards members of out-group (if participant was male then out group was female and if participant was female then out-group was male). Findings revealed that hypothesis H_6 "*there will be difference in ranking of perceived level of social distance across participants' preferred social category in whole sample"* was supported.

Table 3.8

Mean, Standard Deviation and One-Way Analysis of Variance for In-group Identification (IGI), Out-group Identification (OGI) and Social Distance Intensity (SDI) Across Generational Cohorts in their Highly Preferred Social category

Social category	Gen Z		Gen Y		Gen X		Boomers		*F*	*df*	η^2
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*			
Age											
IGI	25.91_a	1.14	24.92_{ab}	1.62	22.56_b	4.95	25.90_a	1.52	3.46*	3,38	.21
OGI	19.36_a	5.20	16.08_a	5.42	17.11_a	2.71	15.30_a	4.27	1.56	3,38	.10
SDI	46.09_a	10.48	42.58_{ab}	14.35	29.78_c	2.68	31.00_{bc}	4.45	7.29***	3,38	.36
Gender											
IGI	26.18_a	2.07	23.87_b	2.75	24.38_a	2.70	24.00_{ab}	1.73	3.21*	3,70	.12
OGI	14.06_a	4.05	14.92_a	3.60	16.38_a	4.94	13.00_a	6.93	1.11	3,70	.04
SDI	35.35_a	8.50	34.55_a	5.74	37.50_a	7.64	38.67_a	9.24	0.88	3,70	.04
Lingual											
IGI	25.83_a	2.04	26.00_a	2.00	26.75_a	0.89	26.80_a	1.64	0.60	3,32	.05
OGI	9.83_a	3.37	14.65_a	5.02	11.38_a	3.42	9.60_a	1.34	3.36*	3,32	.24
SDI	45.50_a	9.59	44.59_a	22.71	52.38_a	28.90	56.00_a	8.48	0.51	3,32	.04
Religion											
IGI	24.86_a	3.46	25.79_a	2.39	25.70_a	2.75	26.63_a	2.17	2.34	3,168	.04
OGI	8.36_a	4.28	8.70_a	4.86	6.10_b	2.71	6.47_{ab}	4.21	4.7**	3,168	.08
SDI	52.06_b	20.58	60.80_b	23.95	86.70_a	25.82	81.80_a	29.01	18.64***	3,168	.25
Place											
IGI	26.40_a	1.55	27.13_a	1.30	25.61_a	3.76	27.33_a	1.03	1.34	3,50	.07
OGI	14.87_a	5.26	15.47_a	4.05	13.39_a	4.26	14.33_a	3.44	0.65	3,50	.04
SDI	38.80_a	13.13	44.87_a	16.19	49.78_a	12.51	40.33_a	11.50	1.94	3,50	.10
Educational group											
IGI	25.38_a	3.00	25.71_a	2.37	24.22_a	3.64	24.50_a	2.38	1.51	3,176	.02
OGI	13.56_a	5.94	13.55_a	6.18	12.39_a	3.50	18.50_a	5.92	1.20	3,176	.02
SDI	55.16_a	20.72	56.52_a	24.88	45.00_a	19.98	53.00_a	20.92	1.30	3,176	.02
Caste											
IGI	26.60_a	2.32	26.43_a	1.65	26.10_a	1.81	25.81_a	3.43	0.29	3,57	.01
OGI	16.40_a	6.62	11.14_b	4.04	9.00_b	4.46	8.38_b	3.03	7.88***	3,57	.29
SDI	48.00_a	23.90	49.43_a	17.82	62.57_a	23.12	68.94_a	20.94	3.11*	3,57	.14
Economic group											
IGI	21.79_a	6.07	22.76_a	3.10	23.08_a	3.51	23.15_a	3.83	0.43	3,97	.01
OGI	13.86_a	6.67	12.62_a	5.23	10.97_a	4.91	9.46_a	3.91	2.24	3,97	.06
SDI	48.21_a	23.28	41.05_a	10.28	47.24_a	18.24	52.38_a	13.40	2.01	3,97	.06

Note. Means in rows with different subscripts differ at the p = .05 level by Bonferroni (MCT). In-group Identification (IGI), Out-group Identification (OGI), and Social Distance Intensity (SDI).
*p < .05. **p < .01. ***p < .001.

In table 3.8, One-way ANOVA was carried out to see if there was any intergenerational effects exists on in-group identification, out-group identification and social distance intensity among generational cohorts in various social categories preferred by participants as their preferred social category. Findings suggested that intergenerational differences on in-group identification were only found to be significant for participants who preferred age F (3, 38) = 3.46, p = .026, η^2 = .21 with η^2 indicating large effect size and gender F (3, 70) = 15.03, p = .028, η^2 = .12 with η^2 indicating medium effect size, as their preferred social category. On the basis of age, Boomers (M = 25.90, SD = 1.52) and Gen Z (M = 25.91, SD = 1.14) identified equally to their in-group whereas Gen X (M = 22.56, SD = 4.95) identified least with their in-group among these four generational cohorts. On the basis of gender, Gen Z (M = 26.18, SD = 2.07) and Gen X (M = 24.38, SD = 2.70) identified themselves equally to their own gender group whereas Gen Y (M = 23.87, SD = 2.75) identified least with their own gender group as reflected by Bonferroni post-hoc analysis.

Intergenerational differences on out-group identification were found to be significant for participants who preferred lingual F (3, 32) = 3.36, p = .031, η^2 = .24 with η^2 indicating large effect size; religion F (3, 168) = 4.71, p = .003, η^2 = .08 with η^2 indicating medium effect size; and caste F (3, 57) = 7.88, p < .001, η^2 = .29 with η^2 indicating large effect size, as their preferred social category. Further, Bonferroni post-hoc analysis revealed that on the basis of language, all generational cohorts identified statistically same to their out-group members but their means indicated that Gen Y (M = 14.65, SD = 5.02) identified most to their out-groups members among these four generational cohorts. On the basis of religion, Gen Z (M = 8.36, SD = 4.28) and Gen Y (M = 8.70, SD = 4.86) identified to same extent to their out-group members whereas Gen X (M = 6.10, SD = 2.71) identified least to their out-group members, and on the basis of caste Gen Y (M = 11.14, SD = 4.04), Gen X (M = 9.00, SD = 4.46) and Boomers (M = 8.38, SD = 3.03) identified equally to their out-group members but Gen Z (M = 16.40, SD = 6.62) identified significantly higher than others.

We found significant intergenerational differences on social distance intensity for participants who preferred age F (3, 38) = 7.29, p < .001, η^2 = .36 with η^2 indicating large effect size; religion F (3, 168) = 18.64, p < .001, η^2 = .25 with η^2 indicating large effect size; and caste F (3, 57) = 3.11, p = .033, η^2 = .14 with η^2 indicating large effect size, as their preferred social category. Further, Bonferroni post-hoc analysis suggested that on the basis of age, intensity of social distance was emerged to be higher in Gen Z (M = 46.09, SD = 10.48) towards older people whereas Gen X (M = 29.78, SD = 4.04) displayed lower intensity of social distance towards members of young group. On the basis of religion, Gen Z (M = 52.06, SD = 20.58) and Gen Y (M = 60.80, SD = 23.95) displayed equal intensity of social distance but significantly lower than the Gen X (M = 86.70, SD = 25.82) and Boomers (M = 81.80, SD = 29.01) towards members of their out-group, and on the basis of caste, all generational cohorts statistically displayed equal intensity of social distance towards members of their out-group but from means we can infer that Gen X (M = 62.57, SD = 23.12) and Boomers (M = 68.94, SD = 20.94) displayed more than other two generational cohorts. Findings revealed that hypothesis H_7 "*there will be intergenerational effects on in-group identification, out-group identification, and social distance intensity in participants' preferred social category across generational cohorts"* was partially supported.

Table 3.9

Intergenerational Effects on Perceived Social Distance

Social category	Generational cohorts	n	M_{rank}	K-W $\chi^2(3)$	ε^2
Age	Gen Z	11	29.36	28.17***	.69
	Gen Y	12	30.13		
	Gen X	9	11.50		
	Boomers	10	11.50		
	Total	42			
Gender	Gen Z	17	37.53	7.68	.10
	Gen Y	38	36.83		
	Gen X	16	35.53		
	Boomers	3	56.33		
	Total	74			
Lingual	Gen Z	6	13.75	3.46	.10
	Gen Y	17	17.38		
	Gen X	8	21.19		
	Boomers	5	23.70		
	Total	36			
Religion	Gen Z	36	57.86	38.68***	.23
	Gen Y	56	70.05		
	Gen X	50	113.30		
	Boomers	30	106.90		
	Total	172			
Place	Gen Z	15	17.80	14.31**	.27
	Gen Y	15	28.67		
	Gen X	18	36.78		
	Boomers	6	21.00		
	Total	54			
Educational group	Gen Z	81	90.81	1.32	.01
	Gen Y	77	93.03		
	Gen X	18	77.81		
	Boomers	4	92.63		
	Total	180			
Caste	Gen Z	10	16.85	11.98**	.20
	Gen Y	14	27.07		
	Gen X	21	33.48		
	Boomers	16	40.03		
	Total	61			
Economic group	Gen Z	14	50.93	12.72**	.13
	Gen Y	37	40.08		
	Gen X	37	55.00		
	Boomers	13	70.77		
	Total	101			

$^{**}p < .01$. $^{***}p < .001$.

The data were measured on ordinal scale; therefore, in table 3.9 Kruskal-Wallis (K-W) rank sum tests were conducted to see if there was any intergenerational effect exists on perceived social distance towards members of respective out-group in various social categories preferred by participants as their preferred social category. There were no significant difference on perceived social distance in gender ($\chi^2(3, n = 74) = 7.68$, $p = .053$, $\varepsilon^2 = .10$), lingual ($\chi^2(3, n = 36) = 3.46$, $p = .326$, $\varepsilon^2 = .10$), and educational group $\chi^2(3, n = 180) = 1.32$, $p = .724$, $\varepsilon^2 = .01$). This suggested that Gen Z, Gen X, Gen Y and Boomers expressed equal level of perceived social distance with small effect size ($\varepsilon^2 < .13$) on these three (gender, lingual, and education) social category. However, significant intergenerational effects were found on perceived social distance in age ($\chi^2(3, n = 42) = 28.17$, $p < .001$, $\varepsilon^2 = .69$), religion ($\chi^2(3, n = 172) = 38.68$, $p < .001$, $\varepsilon^2 = .23$), place ($\chi^2(3, n = 54) = 14.31$, $p = .003$, $\varepsilon^2 = .27$), caste ($\chi^2(3, n = 61) = 11.98$, $p = .007$, $\varepsilon^2 = .20$) and economic group ($\chi^2(3, n = 101) = 12.72$, $p = .005$, $\varepsilon^2 = .13$). The significant K-W test results in these five (age, religion, place, caste and economic group) categories revealed that Gen Z, Gen X, Gen Y and Boomers expressed varied level of perceived social distance; age and place with large effect size ($\varepsilon^2 \geq .26$), religion, caste and economic group with medium effect size ($.13 \leq \varepsilon^2 < .26$).

In age as preferred social category Gen Z ($M_{rank} = 29.36$) and Gen Y ($M_{rank} = 30.13$) expressed more perceived social distance than Gen X ($M_{rank} = 11.50$) and Boomers ($M_{rank} = 11.50$) towards members of out-group. In terms of religion as preferred social category Gen Z ($M_{rank} = 57.86$) and Gen Y ($M_{rank} = 70.05$) expressed less perceived social distance than Gen X ($M_{rank} = 113.30$) and Boomers ($M_{rank} = 106.90$) towards members of out-group and to some extent perceived social distance increased progressively with age. While talking about place as preferred social category Gen Y ($M_{rank} = 28.67$) and Gen X ($M_{rank} = 36.78$) expressed more perceived social distance than Gen Z ($M_{rank} = 17.80$) and Boomers ($M_{rank} = 21.00$) towards members of out-group. In cast as preferred social category Gen Z ($M_{rank} = 16.85$) and Gen Y ($M_{rank} = 27.07$) expressed less perceived social distance than Gen X ($M_{rank} = 33.48$) and Boomers ($M_{rank} = 40.03$) towards members of out-group and increased progressively with age. In economic group Boomers ($M_{rank} = 70.77$) expressed highest perceived social distance towards members of out-group, Gen Z ($M_{rank} = 50.93$) and Gen X ($M_{rank} = 55.00$) almost remained equal whereas Gen Y ($M_{rank} =$

40.08) expressed least among them. Based on results hypothesis H_8 "*there will be intergenerational effects on perceived level of social distance in participants' preferred social category across generational cohorts*" was partially supported.

Table 3.10

Patterns of Ranking of Social Categories by Generational Cohorts

Category	Gen Z	Gen Y	Gen X	Boomers
	M_{rank}	M_{rank}	M_{rank}	M_{rank}
Age	5.12	5.26	5.36	5.02
Gender	4.54	4.63	5.14	5.82
Lingual	5.33	4.87	5.11	4.57
Religion	3.73	3.84	2.76	2.32
Place	4.96	4.97	5.01	5.20
Educational group	2.59	3.37	4.91	5.48
Caste	5.77	5.29	4.70	3.92
Economic group	3.96	3.77	3.01	3.67
n	190	266	177	87
Friedman χ^2 (7)	235.24***	169.01***	213.80***	133.93***
ES (*W*)	.18	.09	.17	.22

Note. N =720. Smaller mean rank indicates higher preference of that social category in each generational cohort.
*** $p < .001$.

The nature of data was ordinal hence in table 3.10, Friedman rank sum non-parametric statistical test was employed to examine whether the medians of social categories were unequal in each generational cohort so that we can identify the patterns of ranking of social categories in each generational cohort. In case of Gen Z the results of the Friedman test were significant, χ^2 (7, n = 190) = 235.24, $p < .001$, W = .18. The value of W indicated small effect size. From the pattern of ranking of social categories it can be inferred that highly preferred social category among Gen Z was educational group (M_{rank} = 2.59) and least preferred social category was caste (M_{rank} = 5.77).

For Gen Y the output of the Friedman test were significant, χ^2 (7, n = 266) = 169.01, $p < .001$, W = .09. The value of W indicated small effect size. From the pattern of ranking of social categories it can be inferred that highly preferred social category among Gen Y was educational group (M_{rank} = 3.37) and least preferred social category was caste (M_{rank} = 5.29).

Moving to next generational cohort Gen X, the findings of the Friedman test were significant, χ^2 (7, n = 177) = 213.80, $p < .001$, W = .17. The value of W indicated small effect size. From the pattern of ranking of social categories it can be inferred that highly preferred social category among Gen X was religion (*M* rank = 2.76) and least preferred social category was age (*M* rank = 5.82).

In the last generational cohort Boomers, the results of the Friedman test were significant, χ^2 (7, *n* = 87) = 133.93, *p* < .001, *W* = .22. The value of *W* indicated small effect size. From the pattern of ranking of social categories it can be inferred that highly preferred social category among Boomers was religion (*M* rank = 2.32) and least preferred social category was gender (*M* rank = 5.36). Findings revealed that hypothesis H_9 "*the ranking of different social categories will vary on perceived social distance in generational cohorts*" was supported.

3.2.3 Aim 4: Category Salience and Intergroup Relationship in Cross-Categorization Situation

Table 3.11

Mean, Standard Deviation and Repeated Measures Analysis of Variance for Social Identification When Religion Dimensions Crossed With Caste Dimensions Resulted in Religion-Caste Cross-Categorization Condition

Sample characteristics	Religion and cast both in-group (ii)		Religion in-group caste out-group (io)		Religion out-group caste in-group (oi)		Religion and cast both out-group (oo)		*F*(*df*)	η_p^2
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*		
Overall	6.00_a	0.98	5.02_b	1.46	3.26_c	1.71	2.56_d	1.48	1087.70*** (3, 2157)	.60
Religious unreserved	6.0_a	0.94	4.94_b	1.56	3.22_c	1.75	2.30_d	1.51	537.56*** (3, 1056)	.60
Religious reserved	6.13_a	0.88	5.50_b	1.09	3.01_c	1.67	2.77_d	1.43	551.07*** (3, 756)	.69
Non-religious unreserved	5.82_a	0.94	4.14_b	1.36	4.02_b	1.52	3.00_c	1.41	80.12*** (3, 147)	.62
Non-religious reserved	5.59_a	1.40	4.25_b	1.56	3.83_b	1.57	2.78_c	1.28	60.60*** (3, 189)	.49

Note. Means in rows with different subscripts differ at the *p* = .05 level by Bonferroni (MCT), ii–religion and caste both in-group, io–religion in-group and caste out-group, oi–religion out-group and caste in-group, oo– religion and caste both out-group.
*** *p* < .001

Table 3.11 shows the mean difference in social identification of individuals across the sample and with respect to their double in-group membership (sample characteristics) in the religion-caste cross-categorization condition using repeated measures ANOVA. In the religion-caste cross-categorization condition, participants were required to provide their social identification score toward members representing each group with respect to them. When religion and caste were crossed, four groups emerged: (1) people with the same religious orientation who belonged to the same caste group (ii– religion and caste both in-group), (2) people with the same religious orientation who belonged to a different caste group (io– religion in-group and caste out-group), (3) people with different religious orientation who belonged to the same caste group (oi– religion out-group and caste in-group), and (4) people with the different religious orientation who belonged to the different caste group (oo–religion and caste both out-group).

The results of overall sample suggested that participants shown significant mean differences on their social identification across religion-caste cross-categorization condition $F(3, 2157) = 1087.70$, $MSE = 1.65$, $p < .001$, $\eta_p^2 = .60$ with large effect size. Bonfferroni post-hoc test revealed that overlapping religion-caste group membership io ($M = 5.02$, $SD = 1.46$) and oi ($M = 3.26$, $SD = 1.71$) remained significant at the .05 level of significance in terms of their social identification. Consequently, we did not observed additive pattern (ii > io = oi > oo) in overall sample across religion-caste cross-categorization condition and religion emerged as salient social category to socially identify with other groups and also as a dimension of social categorization when compared to caste.

When participants were religious and from unreserved category, the mean difference on their social identification was found to be significant across religion-caste cross-categorization condition $F(3, 1056) = 537.56$, $MSE = 1.82$, $p < .001$, $\eta_p^2 = .60$ with large effect size, and post-hoc test suggested that in this group additive pattern was not observed (ii > io = oi > oo), as well as religion emerged as salient social category than caste (io > oi) to socially identify with other groups. We found similar findings for participants who were religious and reserved.

For non-religious and unreserved participants, the mean difference on their social identification was also found to be significant across religion-caste cross-categorization condition $F(3, 147) = 80.12$, $MSE = 0.85$, $p < .001$, $\eta_p^2 = .62$ with large effect size, and post-hoc test suggested that in this group additive pattern was observed (ii > io = oi > oo). Hence, for this group no category salience was reflected between religion and caste (io = oi) which means that they equally favoured religion and caste to socially identify with other groups. We found similar findings for participants who were non-religious and reserved. Based on findings hypothesis H_{10} "*there will be mean differences on social identification among whole sample and participants' double in-group membership in religion-caste cross-categorization condition*" was supported.

Table 3.12

Mean, Standard Deviation and Repeated Measures Analysis of Variance for Social Identification When Age Dimensions Crossed With Gender Dimensions Resulted in Age-Gender Cross-Categorization Condition

Sample characteristics	Age and gender both in-group (ii)		Age in-group gender out-group (io)		Age out-group gender in-group (oi)		Age and gender both out-group (oo)		$F(df)$	η_p^2
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*		
Overall	6.46_a	0.85	5.40_b	1.58	5.49_b	1.42	4.49_c	1.81	336.96*** (3, 2157)	.32
Young male	6.30_a	0.89	5.20_b	1.68	5.24_b	1.54	4.32_c	1.91	89.85*** (3, 717)	.27
Young female	6.51_a	0.86	5.24_c	1.57	5.76_b	1.38	4.36_d	1.86	142.26*** (3, 720)	.37
Old male	6.50_a	0.86	5.65_b	1.61	5.15_c	1.35	4.38_d	1.77	95.80*** (3, 396)	.42
Old female	6.62_a	0.62	5.90_b	1.17	5.86_b	1.07	5.28_c	1.24	46.10*** (3, 315)	.30

Note. Means in rows with different subscripts differ at the $p = .05$ level by Bonferroni (MCT), ii–age and gender both in-group, io–age in-group and gender out-group, oi–age out-group and gender in-group, oo– age and gender both out-group.
*** $p < .001$

Table 3.12 presents the mean differences in social identification of individuals across the sample and with respect to their double in-group membership (sample characteristics) in the age-gender cross-categorization condition. In an age-gender cross-categorization setting, participants were required to give their social identification score to members representing each group with regard to them. When age and gender were crossed, four groups resulted: (1) people of the same age group and gender (ii– age and gender both in-group), (2) people of the same age group but from the opposite gender (io– age in-group and gender out-group), (3) people of different age groups but from the same gender (oi– age out-group and gender in-group), and (4) people of different age groups and representing opposite gender (oo– age and gender both out-group).

The overall sample resulted in a significant mean difference in social identification across age-gender cross-categorization condition $F(3, 2157) = 336.96$, $MSE = 1.38$, $p < .001$, $\eta_p^2 = .32$ with large effect size. Regarding social identification, the Bonfferroni post-hoc test demonstrated that overlapping age-gender group membership io ($M = 5.40$, $SD = 1.58$) and oi ($M = 5.49$, $SD = 1.42$) remained insignificant at the.05 level of significance. As a result, across age-gender cross-categorization conditions, we observed an additive pattern (ii > io = oi > oo) in the whole sample. Therefore, no category salience was reflected between ages and genders across the sample (io = oi), which indicates that people gave equal weight to their ages and genders when attempting to socially identify with other groups. Table 3.12 showed a similar additive pattern when participants having double in-group membership were young males and older females. Gender (io < oi) appeared to be a significant salient social category in young females, whereas age (io > oi) appeared as a significant social category in old males on their social identification towards other groups. Results revealed that hypothesis H_{11} "*there will be mean differences on social identification among whole sample and participants' double in-group membership in age-gender cross-categorization condition*" was supported.

Table 3.13 presents the results of a repeated measure ANOVA to examine the mean difference in participants' social identification across samples and in relation to their double in-group membership (sample characteristics) in the context of the economy-education cross-categorization condition. In a condition involving the cross-categorization of economy and education, participants were required to provide their social identification score toward individuals who represented each group in regard to them. When the economy was crossed with education, it created four groups: (1) people from the same economic and educational group (ii– economy and education both in-group), (2) people from the same economic group but from a different educational group (io– economy in-group and education out-group), (3) people from a different economic group but from the same educational group (oi– economy out-group and education in-group), and (4) people from a different economic and educational group (oo– economy and education both out-group).

Table 3.13

Mean, Standard Deviation and Repeated Measures Analysis of Variance for Social Identification When Economic Dimensions Crossed With Education Dimensions Resulted in Economy-Education Cross-Categorization Condition

Sample characteristics	Economy and education both in-group (ii)		Economy in-group education out-group (io)		Economy out-group education in-group (oi)		Economy and education both out-group (oo)		$F(df)$	η_p^2
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*		
Overall	6.22_a	0.94	4.53_b	1.68	4.60_b	1.84	3.57_c	1.67	412.12*** (3, 2157)	.36
Economically weak and illiterate	6.24_a	0.82	5.08_b	1.34	3.39_c	1.88	2.59_d	1.48	291.69*** (3, 549)	.61
Economically weak and literate	6.31_a	0.88	5.16_b	1.37	4.34_c	1.91	3.26_d	1.53	189.88*** (3, 606)	.49
Economically strong and illiterate	5.99_a	1.02	4.89_{bc}	1.61	5.21_b	1.54	4.69_c	1.41	16.62*** (3, 249)	.17
Economically strong and literate	6.22_a	1.03	3.50_d	1.67	5.49_b	1.19	4.17_c	1.54	277.27*** (3, 744)	.53

Note. Means in rows with different subscripts differ at the *p* =.05 level by Bonferroni (MCT), ii–economy and education both in-group, io–economy in-group and education out-group, oi– economy out-group and education in-group, oo– economy and education both out-group.
*** $p < .001$

With regard to the economy-education cross-categorization condition, the sample exhibited a significant mean difference in social identification $F(3, 2157) = 412.12$, $MSE = 2.12$, $p < .001$, $\eta_p^2 = .36$ with a large effect size. The overlapping economy-education group membership categories io ($M = 4.53$, $SD = 1.68$) and oi ($M = 4.60$, $SD = 1.84$) for social identification remained non-significant at the .05 level of statistical significance, according to the results of the Bonferroni post-hoc test. As a result, we noticed an additive pattern (ii > io = oi > oo) across economy-education cross-categorization conditions in the overall sample. Consequently, there was no category salience between the economic and educational categories throughout the overall sample (io = oi), demonstrating that participants assigned their economic situation and degree of education the same weight when attempting to identify with other groups but this finding was not observed when we assessed participants' on their economy-education double in-group membership.

As suggested by Table 3.13, in participants whose double in-group membership was economically weak and illiterate, their economic identification (io > oi) emerged as significant salient in their social identification towards other groups compared to their educational identification. A similar trend was found in participants with economically weak and literate double in-group membership across economy-education cross-categorization conditions. Participants who belonged to economically strong and literate double in-group membership favoured educational identification (io < oi) than economic identification across economy-education cross-categorization conditions, whereas this phenomenon remained unclear and needs further investigation with participants belonging to the economically strong and illiterate group as their double in-group membership additionally from table 3.13 it appear that they partially followed the baseline pattern. Results revealed that hypothesis H_{12} "*there will be mean differences on social identification among whole sample and participants' double in-group membership in economy-education cross-categorization condition"* was supported.

Table 3.14

Mean, Standard Deviation and Repeated Measures Analysis of Variance for Social Identification When Language Dimensions Crossed With Place Dimensions Resulted in Language-Place Cross-Categorization Condition

Sample characteristics	Language and place both in-group (ii)		Language in-group place out-group (io)		language out-group place in-group (oi)		Language and place both out-group (oo)		F (3, 2157)	η_p^2
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*		
Overall	6.45_a	0.76	5.24_b	1.37	4.40_c	1.76	2.59_d	1.55	1348.53***	.65

Note. Means in rows with different subscripts differ at the p =.05 level by Bonferroni (MCT), ii– language and place both in-group, io–language in-group and place out-group, oi– language out-group and place in-group, oo– language and place both out-group. Here only one cross-categorization occurred because language and place was kept as constant response for each individual.

*** $p < .001$

Repeated measure analysis of variance was used because all participants were needed to provide their social identification score towards members representing each group with respect to them in language-place cross-categorization situation. When language was crossed with place of belonging it resulted in four groups: (1) people speaking native language and were belonging to native place (ii– language and place in-group) (2) people speaking native language and were belonging to different place (io– language in-group and place out-group) (3) people speaking other languages and were belonging to native place (oi– language out-group and place in-group) and (4) people speaking other languages and were belonging to different place (oo– language and place out-group). The results indicated significant mean differences in overall participants' social identification across language-place cross-categorization condition $F(3, 2157) = 1348.53$, $MSE = 1.41$, $p < .001$, $\eta_p^2 = .65$ with large effect size. Furthermore, the results of overall sample in the language-place cross-categorization condition revealed that overlapping language-place group membership io ($M = 5.24$, $SD = 1.37$) and oi ($M = 4.40$, $SD = 1.76$) found to be significant at the .05 level of significance in terms of their social identification, as suggested by the Bonfferroni test. As a result, the language-place cross-categorization condition did not follow the additive pattern (ii > io = oi > oo), and lingual identification emerged as a salient social category for social identification and also as a dimension of social categorization when compared to participants' place of belonging (regional group identification). Based on results, hypothesis H_{13} "*there will be mean differences on social identification among whole sample in language-place cross-categorization condition*" was supported.